6.

Dark Energy

As if you didn't have enough to worry about, the universe in recent decades was discovered to wield a mysterious pressure that issues forth from the vacuum of space and that acts opposite cosmic gravity. Not only that, this "negative gravity" will ultimately win the tug-of-war, as it forces the cosmic expansion to accelerate exponentially into the future.

For the most mind-warping ideas of twentieth-century physics, just blame Einstein.

Albert Einstein hardly ever set foot in the laboratory; he didn't test phenomena or use elaborate equipment. He was a theorist who perfected the "thought experiment," in which you engage nature through your imagination, by inventing a situation or model and then working out the consequences of some physical principle. In Germany before World War II, laboratory-based physics far outranked theoretical physics in the minds of most Aryan scientists. Jewish physicists were all relegated to the lowly theorists' sandbox and left to fend for themselves. And what a sandbox that would become.

As was the case for Einstein, if a physicist's model intends to represent the entire universe, then manipulating the model should be tantamount to manipulating the universe itself. Observers and experimentalists can then go out and look for the phenomena predicted by that model. If the model is flawed, or if the theorists make a mistake in their calculations, the observers will uncover a mismatch between the model's predictions and the way things happen in the real universe. That's the first cue for a theorist to return to the proverbial drawing board, by either adjusting the old model or creating a new one.